- Many-to-Many (via junction tables):
    - The combination of city_id and offence_id in tables like citywise_ipc_stat represents a many-to-many relationship between cities and offences.

Normalization & Design Principles

- 3NF Applied: All composite or multi-valued attributes are split into separate tables (e.g., Offence, City, State)
- Atomicity Preserved: All fields contain indivisible values
- Referential Integrity: Maintained using foreign keys between linked entities
- Scalability: The structure supports future expansion for new metrics or crime types

Question 3:

1. The safest states and cities as determined from the database in terms of crime rates were: Lakshawadeep, Ladakh, anadaman & Nicoba (the UTs) and states like Sikkim, Mizoram and Goa. Safest cities came out to be Coimbatore, Surat and Kozhikode.

2. Crime Rate Analysis for Metro Cities (2022 & Trends)

Highest Overall Crime in 2022 *(from city_offence_stat)*

These cities recorded the highest total crime count in the year 2022:

City	City ID	Crime Count
Delhi	5	2,816,804.80
Mumbai	15	548,809.80
Bengaluru	2	228,806.90
Jaipur	9	220,703.50
Hyderabad	7	165,293.10
Indore	8	146,927.60
Ahmedabad	1	124,576.60
Chennai	3	118,428.60
Lucknow	14	112,989.50

Patna	17	109,673.00

Lowest Overall Crime in 2022 *(from city_offence_stat)*

City	City ID	Crime Count
Coimbatore	4	39,535.40
Surat	19	58,942.80
Kozhikode	13	67,242.70
Kanpur	10	72,310.00
Ghaziabad	6	80,561.30
Kolkata	12	85,756.00
Pune	18	89,938.50
Nagpur	16	96,455.70
Kochi	11	105,775.50
Patna	17	109,673.00

Trends in Violent Crimes (2020–2022)

Based on crimeheadwise_violent_metro_crime_stat using trend (avg yearly change)

Cities with Increasing Violent Crime Trends

City	City ID	Avg Yearly Change
Delhi	5	+754.00
Pune	18	+548.50
Mumbai	15	+418.00
Jaipur	9	+258.00
Kanpur	10	+189.00
Indore	8	+184.50

Hyderabad	7	+159.50
Ahmedabad	1	+129.00
Ghaziabad	6	+123.00
Patna	17	+115.50

Cities with Decreasing Violent Crime Trends

City	City ID	Avg Yearly Change
Chennai	3	-204.00
Lucknow	14	-136.50
Nagpur	16	+25.00
Kozhikode	13	+35.00
Kolkata	12	+45.50
Surat	19	+81.50
Coimbatore	4	+91.50
Kochi	11	+92.00
Bengaluru	2	+112.00
Patna	17	+115.50

Based on city-level data from 2022 and violent crime trends from 2020 to 2022, **Delhi** consistently ranks highest in both overall crime and violent crime trends. On the other hand, cities like **Coimbatore, Surat, and Kozhikode** show relatively low crime counts and minimal upward trends. Notably, cities like **Chennai and Lucknow** show negative trends in violent crime.

States with Highest Rising Crime Trends

- IPC Crimes: States like Delhi, Uttar Pradesh, Rajasthan, and Kerala are seeing consistently rising crime counts.
- SLL Crimes: State Bihar and Gujarat show significant growth in SLL violations.
- Violent Crimes: States like Odisha, Maharashtra and Madhya Pradesh shows a huge spike in violent cases.

These might indicate growing law & order challenges or better reporting mechanisms.

States with Decreasing Crime Trends

- IPC: State Tamil nadu (massive drop), and Gujarat, Assam, Andhra Pradesh — these are seeing shrinking crime counts, which might reflect improved policing or underreporting.
- SLL: Again, Tamil Nadu and Kerala are seeing the largest decline.
- Violent Crimes: Tamil nadu stands out across multiple categories with a drastic downward trend.

3. Analyze the correlation between crime rates of states and cities.

The primary goal is to analyze whether cities with high crime rates are located in states that also exhibit high crime rates. I am specifically comparing:

- State-level Crime Rates (IPC, Violent Crimes, SLL)
- City-level Crime Rates (IPC, Violent Crimes)

Findings and Observations:

High Crime States and Cities:

- Delhi (City) and Delhi (State):
    - City IPC: 2,816,804.80
    - City Violent Crimes: 11,369
    - State IPC: 1,424.10
    - State Violent Crimes: 54.4
    - Trend: Delhi, both as a state and city, has one of the highest crime rates. The city's high IPC and violent crime rate are aligned with the state's crime data, showing a positive correlation.
- Mumbai (City) and Maharashtra (State):
    - City IPC: 548,809.80
    - City Violent Crimes: 4,987
    - State IPC: 148.80
    - State Violent Crimes: 28.4
    - Trend: Mumbai also aligns with the state of Maharashtra in terms of high crime rates, indicating a strong correlation between city and state crime rates.
- Bengaluru (City) and Karnataka (State):
    - City IPC: 228,806.90
    - City Violent Crimes: 2,769
    - State IPC: 167.10

    - State Violent Crimes: 40.3
    - Trend: Bengaluru follows a similar trend, where its high IPC and violent crime rates align with the state's figures.

Moderate Crime States and Cities:

- Hyderabad (City) and Telangana (State):
    - City IPC: 165,293.10
    - City Violent Crimes: 1,364
    - State IPC: 172.80
    - State Violent Crimes: 25.2
    - Trend: Hyderabad is consistent with the state of Telangana in terms of its moderate crime rates.
- Jaipur (City) and Rajasthan (State):
    - City IPC: 220,703.50
    - City Violent Crimes: 1,849
    - State IPC: 418.60
    - State Violent Crimes: 44.9
    - Trend: Jaipur's crime rates show a strong alignment with the state of Rajasthan, indicating a consistent crime trend across both city and state.

Low Crime States and Cities:

- Coimbatore (City) and Tamil Nadu (State):
    - City IPC: 39,535.40
    - City Violent Crimes: 347
    - State IPC: 246.50
    - State Violent Crimes: 29.5
    - Trend: Coimbatore's crime rates are notably low, and this is reflected in Tamil Nadu's state-level data as well.

General Analysis of Correlation:

- Positive Correlation: Cities with high crime rates (such as Delhi, Mumbai, Bengaluru) tend to be in states with similarly high crime rates. This suggests a positive correlation where urban crime rates reflect the overall state crime trends.
- Exceptions: Some cities, like Coimbatore, show low crime rates despite being in states with moderate crime rates, highlighting that not all cities follow the exact pattern of their state.

4. Compare crime patterns across states for cases registered under the Indian Penal Code (IPC) and Special & Local Laws (SLL).

Comparing crime patterns across states for the cases registered under the IPC and SSL we see that there is a positive relationship between the cases registered under IPC and SLL. The correlation for chargesheeting rates between the two across all states is about 0.58 giving moderate relations, that states having higher rates of IPC cases that are chargesheeted likely also have higher rates of SLL crime cases chargesheeted.

At the same time, correlation between cases under IPC and SLL have a correlation of 0.37 which is weak to moderate entailing that states with higher rates of crime tend to have higher rates of SLL crimes as well.

Specifically:

- State Haryana: Both IPC and SLL cases are high, with a crime rate of 418.6.
- State Kerala: Both IPC and SLL cases are notably high with a crime rate of 661.0.
- State Bihar: The IPC rate is 168.1, and the SLL rate is close at 136.8.
- State Punjab: The IPC rate is 142.9, and the SLL rate is 142.9 as well.

5. Investigate the impact of population size on crime rates and discuss how population should be accounted for when interpreting crime data

When analyzing crime statistics, it's essential to understand that raw crime counts (total number of reported cases) do not account for population size. Larger states naturally report more cases simply because more people live there — this does not necessarily indicate a higher crime prevalence.

For example:

- Uttar Pradesh reported 4,01,787 IPC cases in 2022, while
- Delhi reported 3,00,429 IPC cases.

At first glance, UP seems more crime-prone. However, when adjusted for population size:

- UP's crime rate (per 1 lakh population): 171.6
- Delhi's crime rate: 1424.1
  This reveals that Delhi has a much **higher crime rate**, despite fewer total cases.

- **Odisha** (460.8 lakh pop, 43,566 cases) has the **highest violent crime rate**: **94.5 per lakh** — far above the national average — despite not having the highest absolute number.
- **Uttar Pradesh** (2,340.9 lakh pop, 52,579 cases), while having the most **total violent crimes**, has a **much lower rate**: **22.5 per lakh**, due to its huge population.
- **Small UTs like Chandigarh** or **Tripura** show higher violent crime rates than larger states like **Gujarat**, again illustrating the importance of using population-adjusted rates.

Therefore, using crime rates for analysis gives us:

Fair comparison between high- and low-population regions
Identification of true outliers, like states or UTs with disproportionately high or low crime per person
For instance:

- Kerala and Haryana have moderate population sizes but high crime rates.
- Maharashtra and Bihar have high raw case counts, but relatively low crime rates due to large populations.

To correctly interpret crime data we should therefore:

Always analyze per capita crime rates, not just raw crime counts.
Consider population density and urbanization — high-density UTs like Delhi may exhibit higher rates due to more reporting or social factors.

Population must always be normalized when interpreting crime statistics. Using crime rates — cases per lakh population — allows meaningful cross-state comparisons. Without this, populous states appear over-criminalized and smaller ones over-safe (or vice versa).

6. What Is the Relationship Between Crime Rate and Status of Police-Registered Cases?

There is **no direct or consistent relationship** between crime rates and investigation outcomes — but some **important patterns** emerge:

Observed Relationship:

1. High Crime Rate Often Stretches Investigation Capacity
- States like Delhi and Manipur have very high crime rates.
- Their charge-sheeting rates are low (0.4% for Manipur) and pendency is high (above 84%).
- Interpretation: High volume of cases overwhelms investigation and prosecution systems, reducing efficiency.

2. Moderate Crime Rates Can Support Better Case Disposal
- States like Andhra Pradesh and Odisha have moderate crime rates.
- Their charge-sheeting rates are very high (97–100%), though pendency still remains an issue.
- Interpretation: With manageable caseloads, police can investigate more cases effectively.

3. Low Crime Doesn't Always Mean Efficient Handling
- Some low-crime states (e.g., Meghalaya, Tripura) still show:
- Very low chargesheeting and high pendency.
- Interpretation: Low crime doesn't guarantee good system performance — factors like police infrastructure, staffing, will play a role.

There are also some cases such that:
4. Low Crime + Good Investigation:
- Dadra and Nagar Haveli & Daman and Diu
- Crime Rate: 7.4
- Chargesheeting: 100%
- Pendency: 66.7%
Small caseloads handled efficiently.

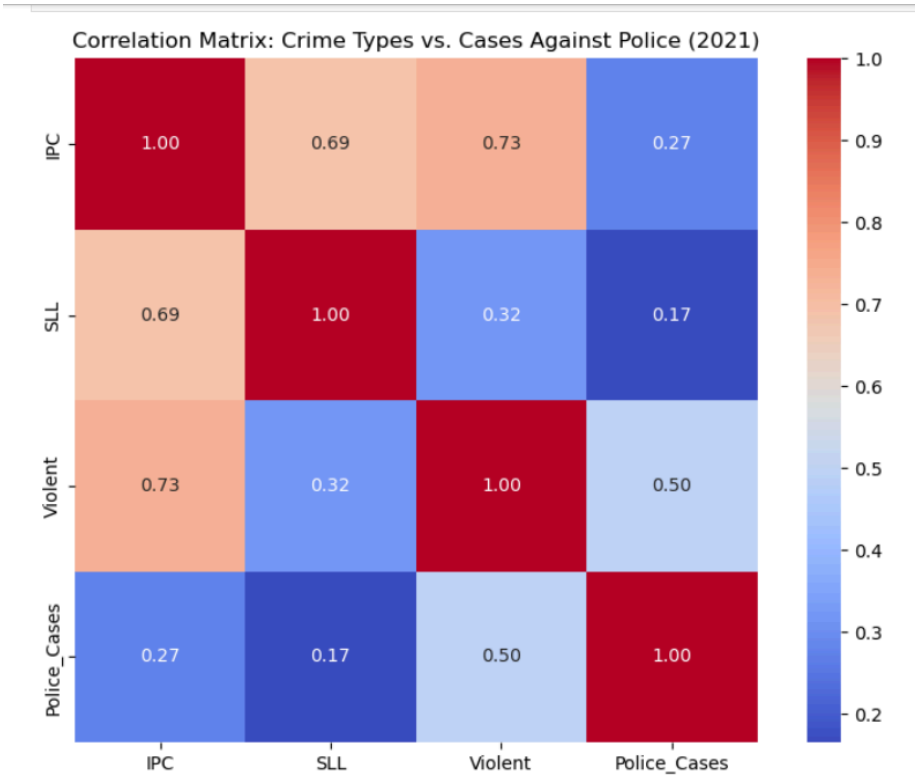
- Sikkim, Uttarakhand
Very low crime, all or most cases disposed with full charge-sheeting.

6. In addition to this, there is no strong or consistent correlation between how high the violent crime rate is and how effectively chargesheets are filed.

- Odisha (94.5 rate) and Uttar Pradesh (22.5) both have very high chargesheeting rates (100% and 93.4%), showing that a high crime rate doesn't necessarily reduce efficiency.
- Manipur, despite a relatively low violent crime rate (19.7), has an extremely poor chargesheeting rate (0.4%), which may suggest systemic issues (lack of resources, backlog, or procedural delays).
- Delhi, with a moderately high violent crime rate (54.4), has only a 69% charge-sheeting rate, showing underperformance.

Interpretation:
Chargesheeting efficiency appears to be more dependent on state-specific capacity and systems rather than crime rate alone.

7. Explore whether there is a connection between crime rates and cases filed against police personnel.



After have created the correlation matrix for crimes across states comparing crime rates and cases against police we can see the following patterns:

Violent ↔ Police Cases	0.50	Moderate positive correlation — more violent crimes may relate to more cases filed against police.
IPC ↔ Police Cases	0.27	Weak correlation — IPC numbers don't strongly explain police cases.
SLL ↔ Police Cases	0.17	Very weak correlation — likely no direct connection.